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Do contextual factors cause differences in passing performance between men and women? A network metrics analysis of the 2022 men's and 2023 women's FIFA World Cups using mixed linear models

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ABSTRACT

This study compared passing network metrics between men's and women's football, considering contextual factors of team/opponent ranking, match stage, and players' position, aiming to understand women's passing performance and how it differs from men's matches. We analysed 128 matches (83,470 passes) from the 2023 Women's World Cup and 2022 World Cup, considering 11 passing metrics (e.g. Degree-based, Centrality-based metrics) and used mixed linear models to identify differences. Results suggest top 16 teams in both cups showed better team cooperation (Decreased Average Weighted Degree, Average Degree, Density, $p < .05$). Women's teams showed less stability than men's when facing top 16 teams (Decreased Average Clustering Coefficient, Average Weighted Degree, Average Degree, Density, $p < .05$). Centre backs in both cups played a key role in team cooperation. (Highest value of Weighted Out-Degree, Weighted In-Degree, Weighted Degree, Out-degree, $p < .05$). Detailed statistical values are reported in the results section. This study offers practical insights for coaches to design representative tasks that enhance stability under pressure for women's teams. It also provides women's baseline data for analysts. Future research should refine positional classifications and use continuous team ranking indicators to better reflect competitive context.

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1. Introduction

Women's football has grown significantly in recent years, yet performance analysis remains largely based on men's data (FIFA, 2022b; Pedersen et al., 2019; Van Lange et al., 2018). This limits our understanding of women's match dynamics and hinders the development of sex-specific training and tactics (Clemente, 2018; Lago et al., 2021; Mclean et al., 2018a; Mikkonen, 2022; Pan et al., 2024; Yu et al., 2020).

Football performance is influenced by various contextual factors that constrain both team and individual player performance during matches, these contextual factors can be thematically categorised into three domains: a) team/opponent ranking, b) match stage and c) player position. These factors play important roles in shaping outcomes in matches (Almeida et al., 2014; Aquino et al., 2019; Arjol-Serrano et al., 2021; Armatas et al., 2009; Clemente et al., 2020; Kubayi & Larkin, 2022; Lago-Peñas et al., 2011; Liu et al., 2019; Magni et al., 2023; Memmert et al., 2019). For team ranking (TR) and ranking of opposition (RO), several studies reported that higher-ranked teams tend to outperform lower-ranked teams in both offensive and defensive aspects. For example, they showed more assists in the Greek First Soccer Division (Armatas et al., 2009), greater ball possession and goal-scoring opportunities in the Chinese Football Super League (Liu et al., 2019), and more intense and organised collective efforts to regain possession in the UEFA Champions League (Almeida et al., 2014). Regarding match stage (MS), findings are mixed. Magni et al. (2023) showed that team performance indicators like total shots, crosses, and shots on target were more frequent in the group stage of the UEFA Champions League (Magni et al., 2023). In contrast, Kubayi and Larkin (2022) found more passes and shots on target at the knock-out stage of the FIFA World Cup 2018 (Kubayi & Larkin, 2022). Player position is also a key contextual factor influencing both team and player performance, as different roles exhibited varying levels of involvement in football passing network (Aquino et al., 2019; Clemente et al., 2020; Pina et al., 2017). For example, central midfielders were the most prominent among all positions due to their significant influence on network connectivity during matches at the FIFA World Cup 2014 (Clemente, Martins, Wong, et al., 2015). In contrast, central defenders exhibited greater prominence at the FIFA World Cup 2022 (Pan et al., 2024).

Going beyond performance outcomes and focusing on the performance process, the study of passing networks (PN) using social network analysis in football has grown significantly to understand team performance, tactics, and player dynamics (Buldú et al., 2019; Clemente, Martins, Kalamaras, et al., 2015; Clemente, Martins, Wong, et al., 2015; Duarte et al., 2012; McLean et al., 2017; Mclean et al., 2018a). According to previous studies, the analysis of passing networks (PN) can be divided into two levels: microscale (player level) and macroscale (team level) (Buldú et al., 2019). Building on this foundation and considering Ramos et al.'s (2017) argument that network analysis in team sports should be closely linked to fundamental sport-specific concepts (e.g. Attacking play), at the team level, passing networks are used to understand the overall structure of a team's play and team passing performance, with metrics like AD (Average Degree), reflecting the overall level of passing involvement, it is based on degree centrality (mean value in a network), which represents the activity level of a player in interactions with teammates and is calculated as the sum of in-degree (number of players who pass to the focal player) and out-degree (number of players to whom the focal player passes); Density describes

the overall level of cooperation or coordination among teammates, higher density values are typically associated with more successful team performance outcomes; ACC (Average Clustering Coefficient), the clustering coefficient (CL) reflects the extent to which a player's teammates are also connected to each other, identifying players involved in tightly connected subgroups versus those less connected. The average clustering coefficient (ACC) represents the mean CL across all players (Buldú et al., 2018; Clemente, 2018; Clemente, Martins, Kalamaras, et al., 2015; McLean et al., 2018b; Pan et al., 2024; Ramos et al., 2017; Ribeiro et al., 2017). At the player level, passing metrics are useful to evaluate individual roles and passing performance (Buldú et al., 2019). Metrics such as Closeness Centrality (CC), which reflects how close a player is to all others in the network, indicating shorter distances to teammates and thus greater potential to quickly receive passes or interact with nearby players; and Eigenvector Centrality (EC), assessing a player's overall influence by considering both direct and indirect connections within the network. Unlike closeness centrality, it accounts for all possible walks from a node, weighted by length, thus capturing how a player impacts not only immediate teammates but also the broader team structure (Ramos et al., 2017; Ribeiro et al., 2017).

Despite the recognised importance of contextual factors and passing networks, few studies have investigated sex differences in other aspects of performance (Almeida et al., 2014; Kubayi & Larkin, 2022; Lago-Peñas et al., 2011; Liu et al., 2019). For instance, Women's teams tend to perform fewer passes with lower accuracy, which has been attributed to differences in technical and physical capacities (Pappalardo et al., 2021). This finding aligns with research comparing men's and women's performances in the UEFA Champions League (Bradley et al., 2014). Additionally, men perform about 30% more high-intensity running than women during matches (Krustrup et al., 2005; Mohr et al., 2008). Nevertheless, women generally show lower physical capacities than men, as revealed by aerobic and anaerobic fitness test (Bangsbo et al., 2014; Bradley et al., 2010, 2012, 2014; Krustrup et al., 2005; Mohr et al., 2008; Mujika et al., 2009).

In summary, although women's football has gained increasing attention in recent years, performance analysis still relies heavily on data derived from men's matches, limiting our understanding of sex-specific match performance. Additionally, while sex-based differences have been reported in football, empirical research investigating sex differences in passing network remains scarce. Therefore, this study aimed to: (1) compare passing network metrics between men's and women's teams; (2) assess the impact of contextual variables such as team ranking, opponent ranking, match stage, and player position; and (3) provide practical and sex-specific insights for coaches and analysts, with implications for the design of training practices. We expect that women's football network metrics are influenced by contextual factors leading to differences in performance when compared with men's performance.

2. Methods

2.1. Sample and data

The sample consisted of all 128 matches (2816 observations of individual players 83,470 passes) played during the FIFA World Cup 2022 ($n = 64$) and the FIFA Women's World Cup 2023 ($n = 64$), involving 32 participating teams in each cup.

The raw data was sourced from the FIFA official website (FIFA, 2022a, 2023a). The FIFA raw data were collected using TRACAB system (New York, U.S.A.). The reliability of tracking system has been validated in the previous research (Linke et al., 2020). To construct an accurate representation of the passing networks in football matches, the raw data were transformed into adjacency matrices. Each matrix consists of nodes representing players and edges representing passes between them. Since each match involves two teams, this process yielded a total of 256 passing matrices. Only 11 players were included in each matrix, with substitutes replacing the players they entered for.

2.2. Procedure

As illustrated in Figure 1, this study's procedure involved the calculation of passing network metrics at both the team (e.g. Density) and player levels (e.g. Eigenvector centrality), utilising mixed linear models to identify differences between men's and women's football under four contextual factors.

2.3. Contextual factors

In this study, four contextual factors were considered: Team ranking (TR), ranking of opposition (RO), match stage (MS), and player position.

This study determined TR and RO based on the most recent FIFA World Cup rankings before the tournaments, following existing methodologies (Lepschy et al., 2020, 2021; Pan et al., 2024). Teams were classified into two groups based on their ranking: Top 16 teams (1st-16th) and bottom 16 teams (17th-32nd) (FIFA, 2022c, 2023b). For the match stage, each match was categorised as either a group stage or a knock-out stage match. To ensure accurate classification of each position, we obtained position data from post-match reports, which were then assessed by two experienced Asian Football Confederation (AFC) coaches (each holding A Level or higher licence) through a detailed review of tactical-view video footage covering the entire match. Two rounds of classification were conducted: the initial classification followed by a reclassification after 30 days. The double-coded position classification and high inter-rater agreement (Cohen's Kappa = 0.91) strengthens the study's internal validity. The final positions included in this study, following the classification method adopted by Pan et al. (2024), were: GK (Goalkeeper, $n = 255$), CB (Centre Back, $n = 576$), FB (Full Back, $n = 440$), CM (Centre Midfielder, $n = 744$), W (Winger, $n = 504$) and CF (Centre Forward, $n = 300$) (Pan et al., 2024).

2.4. Network metrics

A passing matrix reflects the passing relationship among players, in this study, we used a weighted passing matrix, these matrices are based on the number of passes made by two players in the same direction, see Figure 2. To investigate the differences between the men's and women's World Cups, we carefully selected passing network metrics that not only provide insights into changes in the football context but also reflect individual and

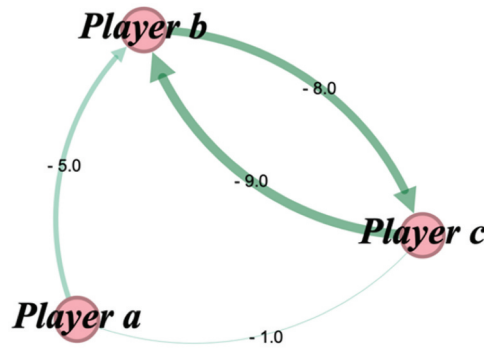


Figure 2. Example of a weighted passing matrix (Gephi Ver 0.10). In a weighted matrix, for example, five passes (edge) were made from player a (node) to player b (node). Thus, the value of the edge is 5.

2.4.1. Team-level metrics

For the team-level, we analysed the following metrics: Average Degree (AD), Average Weighted Degree (AWD), Density, and Average Clustering Coefficient (ACC):

- **AD (Average Degree):** This is the mean of the sum of players' in-degree and out-degree, representing the extent of a player's coordination and cooperation on the pitch. A higher AD indicates better collaboration within the team.
- **AWD (Average Weighted Degree):** This is a variation of AD, which reflects the actual number of passes made and received by a player.
- **Density:** Indicates the overall connectivity of the team. A higher density implies greater team cohesion.
- **ACC (Average Clustering Coefficient):** A higher ACC indicates worse performance due to more frequent passing within specific groups of players.

2.4.2. Player-level metrics

For the player-level, in addition to the team-level metrics, we included the following measures: Closeness Centrality (CC) and Eigenvector Centrality (EC):

- **CC (Closeness Centrality):** Measures how central a player is within the network, indicating their importance in ball circulation and play distribution (Clemente, Martins, Wong, et al., 2015).
- **EC (Eigenvector Centrality):** Similar to CC but also considers the influence of connected players (Cotta et al., 2013).

All metrics were computed using Gephi (Ver 0.10), and the corresponding equations were derived from the official Gephi source code available at: <https://github.com/gephi/gephi/tree/master> (Bastian et al., 2009). The description and interpretations of these formulas were based on prior research (Ramos et al., 2017; Ribeiro et al., 2017). All details are provided in Table 1, G for weighted graph, A for unweighted graph.

Table 1. Description and formula of the network metrics.

Network metrics	Formula and description
Degree	$(1)D_{in}(i) = \sum_{j=1, j \neq i}^N A_{ji} (2)D_{out}(i) = \sum_{j=1, j \neq i}^N A_{ij}$ $(3)D(i) = D_{in}(i) + D_{out}(i)$ In a directed passing network A, the degree of a player is calculated by the sum of the number of players that the player i passes to and receives. The in-degree of player i is measured by the number of players that player i received, and the out-degree of a player is measured by the number of players that player i passes to. N means the number of players in the graph, $i, j \in \{1, 2, \dots, N\}$. Average Degree (AD) is the mean value of degree in a passing network.
Weighted Degree (WD)	$(4)D_{in}^w(i) = \sum_{j=1, j \neq i}^N G_{ji} (5)D_{out}^w(i) = \sum_{j=1, j \neq i}^N G_{ij}$ $(6)D^w(i) = D_{in}^w(i) + D_{out}^w(i)$ In directed passing network G, WD means the total number of passes a player i passes to and receives, focusing on the number of passes. $i, j \in \{1, 2, \dots, N\}$, WID of a player is measured by the number of passes that player i received, WOD of a player is measured by the number of passes that player i passes to. Average Weighted Degree (AWD) is the mean value of weighted degree in a passing network.
Density	$(7)D_s(t) = \frac{E}{N(N-1)}$ For team t, E is the total number of links in the network. N is the number of the players in a network, $N(N-1)$ is the maximum possible number of links in a network, considering each player can connect to $N-1$ other players. In a football passing network, density represents the ratio of actual passes between players to the total possible passes. A higher density means that more players are connected by passes.
Clustering (CL)	$(8)C(i) = \frac{E_i}{k_i(k_i-1)}$ The clustering coefficient $C(i)$ quantifies the extent to which the neighbors of player i are interconnected. In directed passing network A, where k_i is the number of neighbors of player i. The neighborhood $N(i)$ is the set of players connected to i by either an incoming or an outgoing link, and E_i quantifies the number of links within the neighborhood $N(i)$, the symmetrically averaged: $E_i = \sum_{j \neq l, j, l \in N(i)} \frac{(j \rightarrow l) + (l \rightarrow j)}{2}$ Here, $(j \rightarrow l)$ equals 1 if a directed link from node j to node l exists, and 0 otherwise. If both directions exist between a neighbor pair, the pair contributes 1 to E_i; if only one direction exists, it contributes 0.5, $j \neq l, j, l \in N(i)$. In cases $k_i \leq 1$, the clustering coefficient $C_i = 0$. $k_i(k_i - 1)$ represents the maximum possible number of links between neighbors.
Average Clustering Coefficient (ACC)	$(9)A_{(cc)}(t) = \frac{1}{N} \sum_{i \in N'} C(i)$ For team t, $A_{(cc)}(t)$ is the mean of clustering coefficient of all players (N), $C(i)$ represents the clustering coefficient of each player i. N' is the set of players with more than one neighbor, $i \in N'$. This suggests a team tendency to organise into functional clusters, where players preferentially form tightly connected subgroups characterised by high-density interactions.

(Continued)

Table 1. (Continued).

Network metrics	Formula and description
Closeness Centrality (CC)	$(10) C_c^{out}(i) = \frac{ R_i^{out} }{\sum_{j \in R_i^{out}} d_{out}(i,j)}$ For player i, $C_c^{out}(i)$ is out-closeness centrality of player i. In directed passing network A, this metric reflects the value of a given player that is close to the centre of the network. R_i^{out} is number of all players j (where $j \in R_i^{out}$ and $j \neq i$) that are reachable from player i through one or more outgoing directed paths. $\sum_{j \in R_i^{out}} d_{out}(i,j)$ is the sum of the lengths of the shortest directed paths from player i to every other player j. If $R_i^{out} = 0$ (i.e. player i cannot reach any other players in the graph): $C_c^{out}(i) = 0$.
Eigenvector Centrality (EC)	$(11) E_c(i) = \frac{1}{\lambda} \sum_{j=1, j \neq i}^N A_{ji} E_c(j)$ In directed passing network A, eigenvector centrality considers not only the number of connections a player has, but also the importance of the players to whom they are connected. $E_c(i)$ is the eigenvector centrality of player i, and N is the total number of players, j iterates over all other players in the network, with both $i, j \in \{1, 2, \dots, N\}$. The A_{ji} represents the edge from player j to i. λ is the leading eigenvalue of the adjacency matrix A. $E_c(j)$ is the eigenvector centrality score of players j.

2.5. Statistical analysis

The raw data were transformed into passing matrices and imported into Excel for initial preparation. Subsequently, all analyses were conducted using Python (Version 3.10).

To begin with, descriptive statistics for each passing network (PN) variable were generated using the Pandas library (The pandas development team, 2024). Following this, mixed linear models were constructed using the Statsmodels library to analyse differences between the men's and women's World Cup, accounting for the nested structure of the data (i.e. players nested within teams) (Seabold & Perktold, 2010). In these models, PN metrics were set as the dependent variables, while the team was treated as a random effect. At the team-level analysis, team-level PN metrics were set as the dependent variables, team ranking, ranking of opposition, and match stage were included as independent variables, lower ranked teams and group stage teams were set as the baseline. For the player-level analysis, player-level PN metrics were set as the dependent variables, with team ranking, ranking of opposition, and match stage included as independent variables, lower ranked teams and group stage teams were set as the baseline. See equation, $TPNM_{ij}$ is a set of team-level PN metrics, where $\beta_1(TR_{ij}) + \beta_2(RO_{ij}) + \beta_3(MS_{ij})$ were fixed effect of the model, b_i is the random effect (team) and ε_{ij} is the residual error of the model. Model assumptions were tested, no violations of model assumptions (homogeneity, normality) were identified across the models.

$$TPNM_{ij} = \beta_0 + \beta_1(TR_{ij}) + \beta_2(RO_{ij}) + \beta_3(MS_{ij}) + b_i + \varepsilon_{ij}$$

Subsequently, players were categorised into six positional groups, and sex was used as the independent variable for each group to investigate positional differences in PN metrics between men and women, the men's data was set as the baseline. A mixed linear model was constructed for each positional group, with the team treated as a random effect. See the equation, $PPNM_{ij}$ is a set of player-level PN metrics in the same position,

Table 2. Descriptive analysis of network metrics.

Level	Variable	Men (n = 128)		Women (n = 128)	
		Mean	Std	Mean	Std
Team Level	AD	8.246	0.676	6.994	1.041
	AWD	35.709	13.846	27.349	13.309
	Density	0.825	0.068	0.699	0.104
	ACC	0.836	0.066	0.728	0.105
Level	Variable	Men (n = 1408)		Women (n = 1408)	
		Mean	Std	Mean	Std
Player Level	CC	0.863	0.103	0.782	0.122
	CL	0.835	0.078	0.728	0.126
	EC	0.850	0.148	0.776	0.198
	ID	8.251	1.602	6.988	2.014
	OD	8.249	1.518	6.988	1.943
	WID	35.683	21.621	27.335	19.411
	WOD	35.675	23.278	27.335	19.851
	WD	71.358	44.408	54.669	38.577
	Degree	16.501	2.671	13.976	3.523

Table 3. Results from the mixed-linear models for men's and women's team-level passing network metrics.

Variable	Sex	Team	ACC	AD	AWD	Density
				Coef.		
TR	Men	Bottom 16	Baseline			
		Top 16	0.064***	0.607***	14.869***	0.064***
	Women	Bottom 16	Baseline			
		Top 16	0.088***	0.828***	11.703***	0.083***
RO	Men	Bottom 16	Baseline			
		Top 16	-0.011	-0.075	-0.516	-0.008
	Women	Bottom 16	Baseline			
		Top 16	-0.068***	-0.671***	-7.987***	-0.067***
MS	Men	Group	Baseline			
		Knock-out	0.002	0.030	0.626	0.002
	Women	Group	Baseline			
		Knock-out	0.013	0.153	0.406	0.015

TR Team ranking; RO ranking of opposition; MS Match stage. * $p < .05$, ** $p < .01$, *** $p < .001$. AD average degree; AWD average weighted degree; ACC average clustering coefficient.

where $\beta_1(\text{Sex}_{ij})$ is fixed effect of the model, b_i is the random effect (team) and ε_{ij} is the residual error of the model. Model assumptions were tested, no violations of model assumptions (homogeneity, normality) were identified across the models.

$$PPNM_{ij} = \beta_0 + \beta_1(\text{Sex}_{ij}) + b_i + \varepsilon_{ij}$$

We used model-based *Cohen's d* to quantify the effect sizes, providing a measure of the magnitude of differences in the metrics accounting for random effects. Complete numerical values are provided in Supplementary Tables S1-S3. And using the mean difference (MD) to provide a direct measure of the magnitude of differences in metrics. The level of significance was set at $\alpha = 0.05$, with p-values adjusted for multiple comparisons (Benjamini & Hochberg, 1995).

3. Results

Table 2 presents the descriptive analysis of passing network (PN) metrics at both team and player levels for the men's and women's FIFA World Cups. The following

abbreviations are used throughout: AD (average degree), AWD (average weighted degree), ACC (average clustering coefficient), CL (clustering coefficient), ID (indegree), OD (outdegree), WID (weighted indegree), WOD (weighted outdegree), CC (closeness centrality), EC (eigenvector centrality), WD (weighted degree).

3.1. Team PN metrics

The results from the mixed-linear model for men's and women's team-level passing network metrics are summarised in Table 3. The top 16 teams demonstrated higher team connectivity and greater passing volumes than bottom 16 teams in both tournaments. According to the team ranking, Average Clustering Coefficient ($p < .001$, Men-ES = 1.127, Women-ES = 1.187; Men-MD = 0.064, Women-MD = 0.088), Average Degree ($p < .001$, Men-ES = 1.059, Women-ES = 1.161; Men-MD = 0.607, Women-MD = 0.828), Average Weighted Degree ($p < .001$, Men-ES = 1.522, Women-ES = 1.279; Men-MD = 14.869, Women-MD = 11.703), Density ($p < .001$, Men-ES = 1.117, Women-ES = 1.160; Men-MD = 0.064, Women-MD = 0.083) of the top 16 teams were significantly different than the bottom 16 teams in both men and women World Cup. Depending on the ranking of the opposition, only the women's World Cup showed significant differences in Average Clustering Coefficient ($p < .001$, ES = -0.907; MD = -0.068), Average Degree ($p < .001$, ES = -0.940; MD = -0.671), Average Weighted Degree ($p < .001$, ES = -0.873; MD = -7.987), and Density ($p < .001$, ES = -0.940; MD = -0.067). The results showed no significant difference in men's and women's World Cup according to the match stage.

3.2. Player PN metrics

For player-level PN metrics analysis, we considered two dimensions of analysis. In summary, players from the top 16 teams demonstrated higher connectivity and greater passing volumes than from bottom 16 teams in both tournaments. As Table 4 shows, when considering TR, we found that all variables exhibited differences between the top 16 teams and the bottom 16 teams (All ES values showed in the Supplementary Table S2, $p < .01$) in both cups. For RO, Clustering ($p = .005$, ES = -0.197; MD = -0.020), Weighted Degree ($p < .001$, ES = -0.162; MD = -5.367), Weighted In-Degree ($p = .024$, ES = -0.16; MD = -2.678) and Weighted Out-Degree ($p = .03$, ES = -0.154; MD = -2.657) in women are statistically different. Additionally, the stage of the match had an impact on Weighted Degree ($p = .012$, ES = -0.199; MD = -6.584), Weighted In-Degree ($p = .03$, ES = -0.197; MD = -3.286), and Weighted Out-Degree ($p = .018$, ES = -0.189; MD = -3.264) in the women's World Cup.

Table 5 showed the player-level analysis of position, in which sex differences emerged in most of the variables (All ES values showed in the Supplementary Table S3, $p < .05$). In general, centre backs (CB) were the most central players in both tournaments, whereas centre forwards (CF) exhibited the lowest connectivity. Across all positions, men's metrics were generally higher than women's, except for Eigenvector Centrality (CI: -0.043, 0.054), In-Degree (CI: -1.021, 0.016), Out-Degree (CI: -1.005, 0.002), Weighted Degree (CI: -4.003, 2.459), Weighted In-Degree (CI: -4.003, 2.459), and

Table 4. Results from the mixed-linear models for men's and women's player-level passing network metrics.

Variable	Sex	Team	CC	CL	Degree	EC	ID Coef.	OD	WD	WID	WOD
TR	Men	Bottom 16 Baseline									
		Top 16	0.041***	0.058***	1.190***	0.044***	0.591***	0.607***	27.092***	13.655***	13.430***
RO	Women	Bottom 16 Baseline									
		Top 16	0.060***	0.100***	1.928***	0.052**	0.965***	0.965***	26.060***	13.031***	13.036***
	Men	Bottom 16 Baseline									
		Top 16	-0.009	-0.009	-0.174	-0.011	-0.074	-0.092	-1.601	-0.835	-0.747
MS	Women	Bottom 16 Baseline									
		Top 16	-0.010	-0.020**	-0.245	0.005	-0.109	-0.113	-5.367*	-2.678*	-2.657*
	Men	Group Baseline									
		Knock-out	0.002	0.003	0.050	0.004	0.022	0.024	1.752	0.849	0.912
Women		Group Baseline									
		Knock-out	-0.008	-0.013	-0.192	0.002	-0.089	-0.091	-6.584*	-3.286*	-3.264*

TR Team ranking; RO ranking of opposition; MS Match stage. * $p < .05$, ** $p < .01$, *** $p < .001$. Abbreviations: WID weighted indegree; WOD weighted outdegree; CC closeness centrality; EC eigenvector centrality; CL clustering; ID indegree; OD outdegree; WD weighted degree.

Table 5. Results from the models for men's and women's player-level passing network metrics according to the positions.

Position	Sex	CC	CL	Degree	EC	ID Coef.	OD	WD	WID	WOD
GK	Men(Baseline)	0.823	0.879	13.094	0.570	5.369	7.712	39.008	15.482	23.524
	Women	0.680***	0.763***	8.822***	0.405***	3.619***	5.227***	25.789***	9.382***	16.415***
W	Men(Baseline)	0.811	0.841	15.881	0.865	8.349	7.544	53.347	29.817	23.544
	Women	0.739***	0.748***	13.521***	0.819**	7.205***	6.331***	43.558*	24.065**	19.511*
CF	Men(Baseline)	0.764	0.853	14.611	0.819	7.866	6.741	34.207	19.189	15.031
	Women	0.730*	0.743***	13.589*	0.825	7.363	6.239	33.929	18.417	15.538
CB	Men(Baseline)	0.919	0.817	17.415	0.859	8.386	9.031	98.833	47.426	51.409
	Women	0.818***	0.694***	14.271***	0.731***	6.748***	7.531***	73.372**	35.484**	37.893**
FB	Men(Baseline)	0.867	0.823	16.915	0.877	8.578	8.331	74.893	38.691	36.202
	Women	0.768***	0.716***	13.993***	0.806***	7.192***	6.810***	57.935*	30.825*	27.123*
CM	Men(Baseline)	0.891	0.822	17.504	0.910	8.853	8.657	77.366	38.080	39.300
	Women	0.843***	0.707***	15.650***	0.860***	7.696***	7.962***	55.951***	26.514***	29.452***

* $p < .05$, ** $p < .01$, *** $p < .001$. Abbreviations: WID weighted indegree; WOD weighted outdegree; CC closeness centrality; EC eigenvector centrality; ID indegree; OD outdegree; WD weighted degree.

Weighted Out-Degree (CI: $-2.377, 3.390$) in centre forwards (CF), which showed no significant sex differences.

4. Discussion

Overall, the results of this study suggest that when considering team ranking, PN metrics for both men and women show similar changes, but there are other aspects that are different according to sex. To our knowledge, this is the first study to a) investigate how PN metrics of teams and player level in both men's and women's FIFA World Cups changed according to team ranking, ranking of opposition and match stage. b) examine the player-level network metrics in the two tournaments, analysing how they differentiate between men and women players based on their positions.

To guide the discussion, we divided it into two main themes: a) Team-level analysis: comparison of team-level PN metrics according to the team ranking, ranking of opposition and match stage. b) Player-level analysis: comparison of player-level PN metrics according to the team ranking, ranking of opposition and match stage; Sex-based player-level PN metrics comparisons across tactical positions (e.g. centre back, full back).

4.1. Team-level analysis

In the team-level analysis considering the impact of team ranking (TR), as [Table 3](#) shows, the top 16 teams showed higher values in Average Weighted Degree (Men-ES = 1.522, Women-ES = 1.279; Men-MD = 14.869, Women-MD = 11.703), Average Degree (Men-ES = 1.059, Women-ES = 1.161; Men-MD = 0.064, Women-MD = 0.088), and Density (Men-ES = 1.117, Women-ES = 1.160; Men-MD = 0.064, Women-MD = 0.083) in both cups, indicating more frequent passes and receives, stronger connectivity among teammates in the top 16 teams. The analysis of the 2010–2022 FIFA World Cups conducted by Pan et al. (2024), and the study by Aquino et al. (2019) on the 2018 FIFA World Cup, both identified a correlation between possession-based play among higher-ranked teams and enhanced cooperation. These findings are consistent with our observed network characteristics according to TR (Pan et al., 2024; Praça et al., 2019). Building on these previous studies, our results suggest that top 16 women's teams also adopt a possession-based strategy to maintain their dominance, and that the frequent passing between teammates is associated with more scoring opportunities, as it enhances control, disrupts the opponent's defence, and creates space for attack. These characteristics distinguish these top women's teams from the lower ranked teams (Praça et al., 2019; Rampinini et al., 2009). To better understand team-level analysis, we also investigated the changes in player level according to the contextual factors. As [Table 4](#) and [Figure 4](#) shows, Clustering (Men-ES = 0.891, Women-ES = 0.991; Men-MD = 0.058, Women-MD = 0.100), Degree (Men-ES = 0.468, Women-ES = 0.606; Men-MD = 1.190, Women-MD = 1.928), In-degree (Men-ES = 0.380, Women-ES = 0.515; Men-MD = 0.591, Women-MD = 0.965), Out-degree (Men-ES = 0.414, Women-ES = 0.538; Men-MD = 0.607, Women-MD = 0.965), Weighted Degree (Men-ES = 0.679, Women-ES = 0.786; Men-MD = 27.092, Women-MD = 26.060), Weighted In-Degree (Men-ES = 0.709, Women-ES = 0.780; Men-MD = 13.655, Women-MD = 13.031), and Weighted Out-Degree (Men-ES = 0.633, Women-ES = 0.756; Men-MD = 13.430, Women-MD = 13.036) were higher

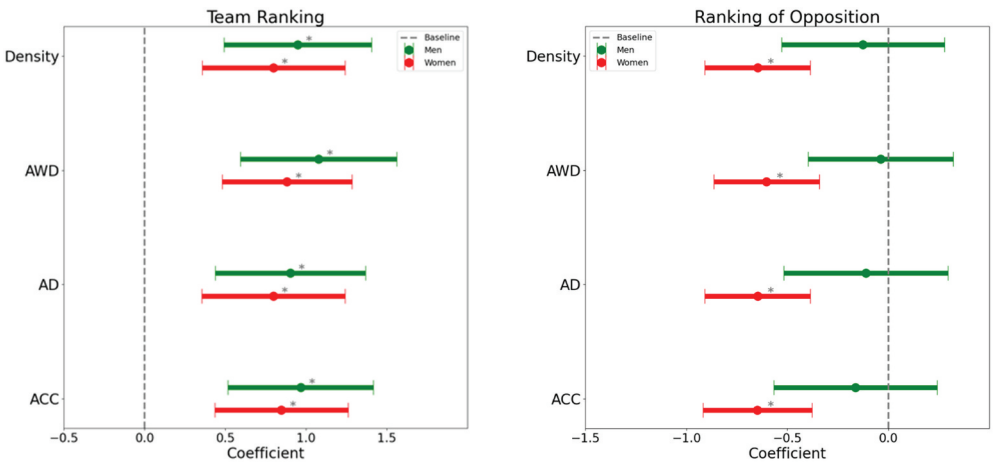


Figure 3. Men and women team-level passing network metrics' tendency compared to the baseline. To facilitate visual comparison across metrics on a common scale, all coefficients were standardised. *indicates $p < 0.05$. Abbreviations: AD average degree; AWD average weighted degree; ACC average clustering coefficient. Baseline: Bottom 16 teams (Team Ranking); Top 16 teams (Ranking of Opposition).

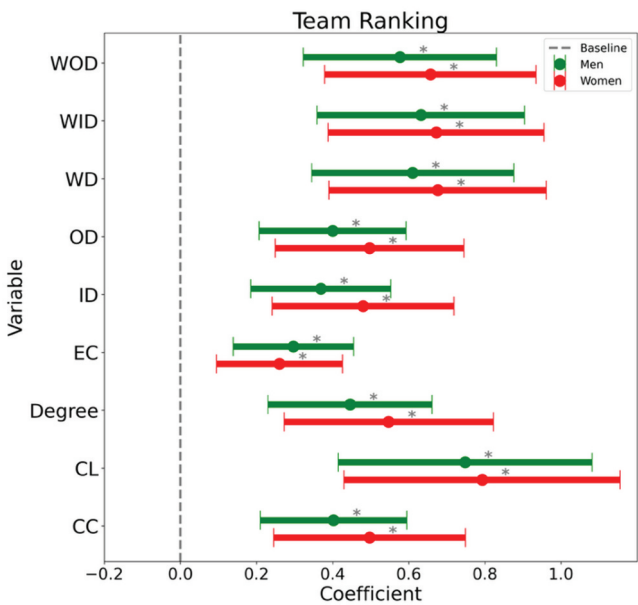


Figure 4. Men and women player-level passing network metrics' tendency compared to the baseline. To facilitate visual comparison across metrics on a common scale, all coefficients were standardised. *indicates $p < 0.05$. Abbreviations: WID weighted indegree; WOD weighted outdegree; CC closeness centrality; EC eigenvector centrality; CL Clustering; ID indegree; OD outdegree; WD weighted degree. Baseline: Bottom 16 teams.

compared to the bottom 16 teams when considering TR. At the player level, top teams also showed that more passes and receptions are associated with greater interconnectivity and cooperation (increased Degree, ID, OD, WD, WID, and WOD), and higher CL suggested that players from the top 16 teams are more likely to form tightly knit passing units with their teammates, reflecting player-level passing advantages. These advantages subsequently scale up to enhance overall team passing performance (Clemente, Martins, Kalamaras, et al., 2015).

According to Table 3 and Figure 3, RO was strongly associated with the women's World Cup. When women's teams faced the top 16 opponents, Average Clustering Coefficient (ES = -0.907; MD = -0.068), Average Weighted Degree (ES = -0.873; MD = -7.987), Average Degree (ES = -0.940; MD = -0.671), and Density (MD = -0.940; MD = -0.067) showed decreases. Women's teams exhibited greater variability in passing network metrics when facing top 16 opponents, possibly due to a wider disparity in quality. In contrast, men's teams performed more consistently across competitive contexts. This pattern may be related to the greater competitive balance observed in men's football, whereas women's football tends to exhibit larger disparities between teams. Consequently, the wider performance gaps among women's teams could lead to greater variability in the data. These observations align with previous studies reporting similar findings in the English, French, German, and Spanish leagues, which indicate that men's football generally has a more balanced competitive structure (Kringstad, 2020; Zamboni-Ferraresi et al., 2018).

4.2. Player-level analysis

The present study also examined the association between TR, RO, and MS and player PN metrics. According to Figure 4, TR was strongly associated with player PN metrics, indicating a large difference between the top 16 teams and bottom 16 teams in both cups. As Table 4 shows, we found Closeness Centrality (Men-ES = 0.416, Women-ES = 0.540; Men-MD = 0.041, Women-MD = 0.060) and Eigenvector Centrality (Men-ES = 0.277, Women-ES = 0.266; Men-MD = 0.044, Women-MD = 0.052) were higher in the top 16 teams compared to the bottom 16 teams that participated in both World Cups. This suggests that a greater number of players in the top 16 teams were centrally positioned within the passing network, reflecting better ball circulation (as indicated by higher Closeness Centrality) and stronger integration within well-connected intra-team passing structures (as indicated by higher Eigenvector Centrality). These findings may reflect a mutually reinforcing relationship between Closeness Centrality and Eigenvector Centrality. Specifically, higher Closeness Centrality indicates that players are more centrally located within the passing network, enabling more efficient ball circulation. Such central positioning not only strengthens passing connections with surrounding teammates but also elevates a player's tactical importance – reflected in higher Eigenvector Centrality, which considers not only the number of connections a player has but also the influence of those they are connected to (Ramos et al., 2017). For the RO analysis, four significant variables were observed: Clustering (ES = -0.197; MD = -0.020), Weighted In-Degree (ES = -0.160; MD = -2.678), Weighted Out-Degree (ES = -0.154; MD = -2.657) and Weighted Degree (ES = -0.162; MD = -5.367), which all showed a decreasing trend when facing a top 16 team in the women's World Cup,

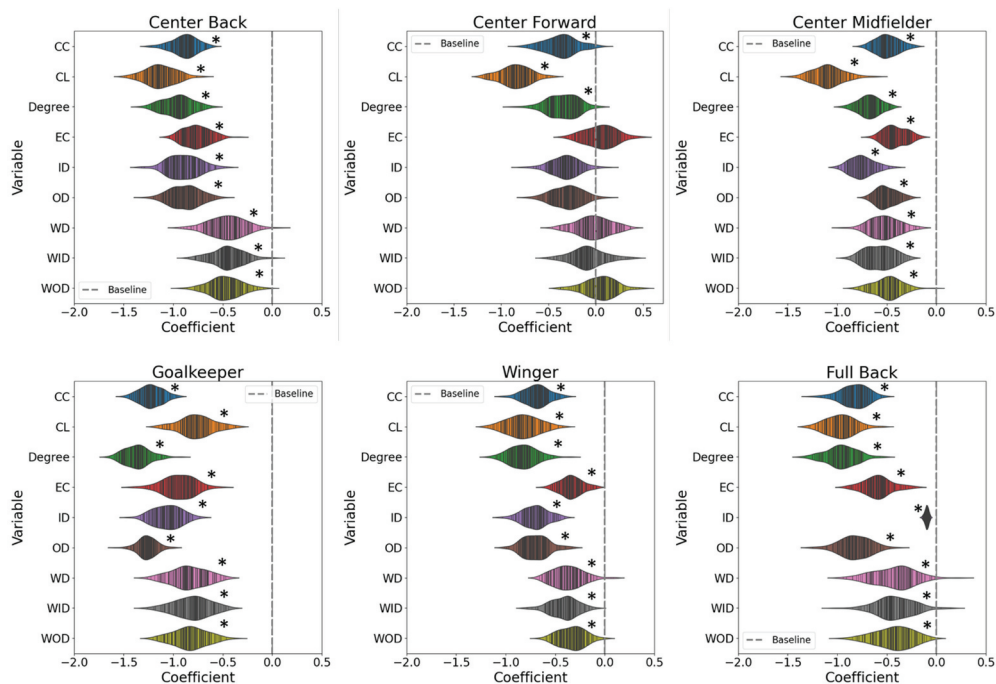


Figure 5. Women player-level passing network metrics' tendency compared to the Men (baseline) according to the positions. To facilitate visual comparison across metrics on a common scale, all coefficients were standardised. *indicates $p < 0.05$. Abbreviations: WID weighted indegree; WOD weighted outdegree; CC closeness centrality; EC eigenvector centrality; CL clustering; ID indegree; OD outdegree; WD weighted degree.

indicating that when facing top 16 teams, players from the bottom 16 teams exhibited fewer passing connections (decreased WD, WID, WOD) and were less likely to form tightly knit passing clusters (decreased CL). According to previous research, this pattern could be related to the possession-based playing style typically adopted by top 16 teams, who tend to dominate ball control and push the game into the opponent's half, as a result, bottom 16 teams have reduced possession time, limiting their ability to maintain structured passing sequences and cohesive sub-group interactions among players (Pan et al., 2024). As for MS on PN metrics, only Weighted Degree ($ES = -0.199$; $MD = -6.584$), Weighted In-Degree ($ES = -0.197$; $MD = -3.286$) and Weighted Out-Degree ($ES = -0.189$; $MD = -3.264$) exhibited differences in women's teams, while no comparable influence was observed in men's teams. This suggests that, compared to the group stage, knockout stage matches in the Women's World Cup featured fewer successful passes and receives. One possible explanation is that greater disparities in individual passing ability may exist between the group and knockout stages in women's tournaments. Previous research has indicated that women's football leagues tend to be more competitively unbalanced than men's leagues, potentially reflecting broader structural disparities that extend to international competitions. Such structural imbalance may lead to more uneven matchups in the group stage, allowing dominant teams to accumulate more successful passes, and thus contribute to the observed decline in passing metrics as

the competition progresses into more balanced knockout rounds. In contrast, the men's tournament showed no such pattern, possibly due to more consistent competitive levels across stages (Kringstad, 2020).

Figure 5 shows the analysis of player-level PN metrics according to the positions. Most metrics in the men's World Cup were higher than those in the women's Cup across all positions, with the exception of Eigenvector Centrality, In-Degree, Out-Degree, Weighted Degree, Weighted In-Degree and Weighted Out-Degree for Centre Forwards, which may indicate generally stronger passing performance among men's players compared to women's players. According to Table 5, both the men's and women's World Cups demonstrated similar patterns, with Centre Backs (CB) showing the highest values in Weighted Out-Degree (Men-mean = 51.409, Women-mean = 37.893, ES = -0.615), Weighted In-Degree (Men-mean = 47.426, Women-mean = 35.484, ES = -0.575), Weighted Degree (Men-mean = 98.833, Women-mean = 73.372, ES = -0.599) and Out-degree (Men-mean = 9.031, Women-mean = 7.531, ES = -1.103). These findings are consistent with CBs occupying a central role in team cooperation and coordination. This aligns with previous studies that identified increasing centrality measures for central defenders, while contrasting with other research emphasising the importance of central midfielders (Aquino et al., 2019; Clemente, Martins, Wong, et al., 2015; Praça et al., 2019). The observed differences might reflect tactical evolution, as our results are based on 2022 and 2023 FIFA World Cup data (consistent with Pan et al.'s study), while other studies used earlier data. The rapid development of football tactics and rules may account for these evolving team performance patterns (Bush et al., 2015; Wallace & Norton, 2014). And furthermore, following the 2018 update by the International Football Association Board (IFAB) to the goal kick rule – where the ball no longer needs to leave the penalty area – teams can more easily initiate build-up play from the defensive zone (FIFA, 2018; Pan et al., 2024). This change likely increases the involvement and importance of centre backs in the early phases of possession. According to Table 5, centre forwards (CF) showed the lowest values in Closeness Centrality, Weighted Out-Degree, Weighted In-Degree, and Weighted Degree (Closeness Centrality, Men-mean = 0.764, Women-mean = 0.730; Weighted Out-Degree, Men-mean = 15.031, Women-mean = 15.538; Weighted In-Degree, Men-mean = 19.189, Women-mean = 18.417; Weighted Degree, Men-mean = 34.207, Women-mean = 33.929, all ES values showed in Supplementary Table S3, $p > 0.05$) among all positions except for goalkeepers (GK) between men and women in both cups. This suggests CFs typically occupy less central network positions with fewer passes, potentially reflecting their specialised role in constantly moving to create space and generate attacking opportunities, as documented in previous studies (Liu et al., 2021; Yi et al., 2019).

4.3. Practical application and limitation

To our knowledge, this is the first study to investigate sex-based differences in passing network metrics across varying match contexts. Our findings indicate that passing networks in the Women's World Cup displayed greater variability against top 16 opponents compared to the men's cup, indicating the need for tactical approaches that enhance stability and adaptability in high-pressure scenarios for

women's teams. Based on network data, coaches should design representative tasks that encourage players to adapt to high-pressing environment while maintaining possession. Moreover, as centre forwards (CFs) had the lowest values in Closeness Centrality, Weighted Out-Degree, Weighted In-Degree, and Weighted Degree in both cups, it is recommended that tactical strategies, based on network data, seek to more actively integrate CFs into the build-up phases of play to enhance their contribution to offensive organisation and overall team connectivity, particularly for women's teams. Furthermore, this study offers valuable insights for coaches in designing training strategies based on women's characteristics and optimising match preparations accordingly. For analysts, this study provides baseline data for women's football, facilitating more effective and context-aware data analysis.

This study has limitations that need to be acknowledged. The position classification did not account for players' attacking or defensive characteristics, which may impact the findings. Additionally, the binary classification of team/opponent ranking (e.g. top 16 vs. bottom 16) may oversimplify the competitive landscape. Future research could consider continuous or fine-grained slots to better reflect variation in team ranking and consider the specific role of the players.

5. Conclusion

Our analysis revealed consistent trends in both World Cups while identifying significant sex-based differences in passing networks. The top 16 teams in both tournaments demonstrated stronger team coordination. Notably, women's teams exhibited greater variability in passing network metrics when competing against top 16 opponents compared to their male counterparts. Positional analysis highlighted the key role of centre backs in team connectivity for both cups, with CF having the lowest values of most passing network metrics in both cups. This study contributes to a growing body of research that supports the need for sex-specific performance analytics in elite football.

Authors contributions

L.X: Conceptualisation, Methodology, Validation, Data curation, Formal analysis, Visualisation, Investigation, Writing – original draft, Writing – review & editing. **C.X:** Validation, Investigation. **G.S:** Methodology, Writing – review & editing. **D.C:** Methodology, Writing – review & editing. **FM.C:** Conceptualisation, Writing – review & editing. **H.B:** Supervision, Conceptualisation, Methodology, Writing – review & editing. **D.A:** Supervision, Conceptualisation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request

Code availability statement

The code used during the current study is available from the corresponding author on reasonable request.

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